Bisection of Chords

Exam code: 9MA0

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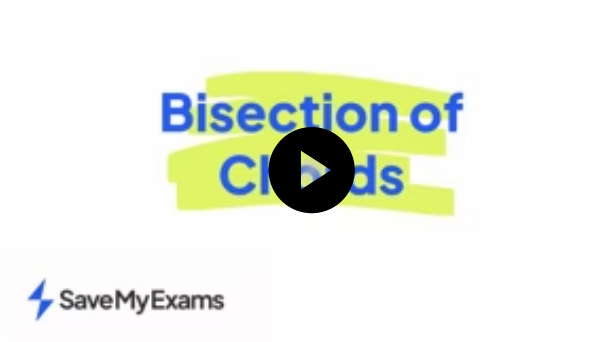

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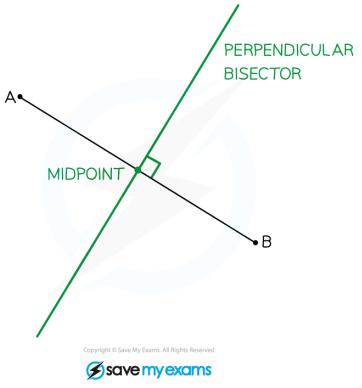
Help me understand

Bisection of chords

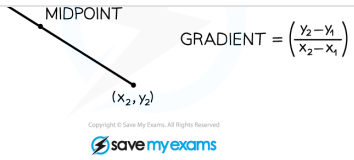
How can I find the equation of a perpendicular bisector?

- The **perpendicular bisector** of a line segment:
  - is perpendicular to the line segment
  - goes through the midpoint of the line segment



- The midpoint and gradient of the line segment between points (x_1, y_1) and (x_2, y_2) are given by the formulae

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- The gradient of the perpendicular bisector is therefore

$$\text{GRADIENT OF PERPENDICULAR BISECTOR} = -\left(\frac{x_2-x_1}{y_2-y_1}\right)$$

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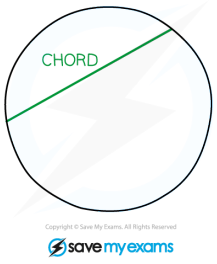
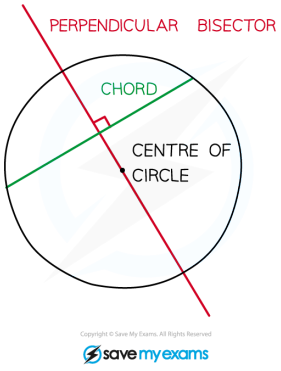
- The equation of the perpendicular bisector is the equation of the line with that gradient through the line segment's midpoint (see Equation of a Straight Line)

How can I use perpendicular bisectors to find the equation of a circle?

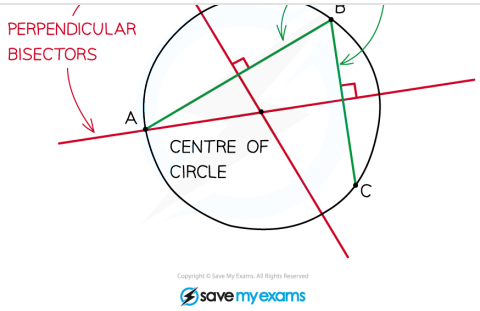
- A **chord** of a circle is a straight line segment between any two points on the circle



- The perpendicular bisector of a chord always goes through the centre of the circle



- If you know three points on a circle, draw any two chords between them – the perpendicular bisectors of the chords will meet at the centre of the circle



- Now that you know the centre of the circle and a point on the circle you can write the equation of the circle

Examiner Tips and Tricks

- To find the point of intersection of two straight lines, set the equations of the lines equal to each other and solve.

Worked Example

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